

Example 4: solved system components

Plant, controller, and augmented-graph components are displayed as PI operators through the six Π_4 kernels $(P, Q_1, Q_2, R_0, R_1, R_2)$. Dynamic filters are displayed as Laplace-domain transfer matrices. Omitted polynomial terms have zero coefficient.

Signal ordering: rows and columns

All vectors below are column vectors. The plant has $n_z = 5$ regulated outputs, $n_w = 3$ exogenous inputs, and $n_u = 1$ control inputs.

$$u = [u_S]^\top, \quad z_\Delta = [z_S, z_G]^\top, \quad w_\Delta = [w_S, w_G]^\top = \delta(z_\Delta), \quad z_p = [x_S, x_G, \mu u_S]^\top.$$

$$z = \begin{bmatrix} z_\Delta \\ z_p \end{bmatrix} = [z_S, z_G, x_S, x_G, \mu u_S]^\top, \quad w = \begin{bmatrix} w_\Delta \\ w_p \end{bmatrix} = [w_S, w_G, w_p]^\top.$$

Primal filter. The columns of Θ are ordered by $[z^\top, w^\top]^\top$; its rows are ordered by $[\tilde{z}^\top, \tilde{w}^\top]^\top$. Explicitly,

$$\begin{bmatrix} \tilde{z}_\Delta \\ \tilde{z}_p \\ \tilde{w}_\Delta \\ \tilde{w}_p \end{bmatrix} = \Theta \begin{bmatrix} z_\Delta \\ z_p \\ w_\Delta \\ w_p \end{bmatrix}, \quad [z^\top, w^\top]^\top = [z_S, z_G, x_S, x_G, \mu u_S, w_S, w_G, w_p]^\top, \quad \text{block sizes } (2, 3, 2, 1).$$

Dual filter. Dual channel positions correspond to $[w^\top, z^\top]^\top$: the dual plant output $\underline{z} \in \mathbb{R}^3$ has the channel order of w , and its input $\underline{w} \in \mathbb{R}^5$ has the channel order of z . Underlined signals belong to the dual system.

$$\begin{bmatrix} \underline{\tilde{z}}_\Delta \\ \underline{\tilde{z}}_p \\ \underline{\tilde{w}}_\Delta \\ \underline{\tilde{w}}_p \end{bmatrix} = \mathbf{D}(\Theta) \begin{bmatrix} \underline{\tilde{z}}_\Delta \\ \underline{\tilde{z}}_p \\ \underline{\tilde{w}}_\Delta \\ \underline{\tilde{w}}_p \end{bmatrix}, \quad \text{channel positions } [w_S, w_G, w_p, z_S, z_G, x_S, x_G, \mu u_S]^\top, \quad \text{block sizes } (2, 1, 2, 3).$$

The scalar factors H_i, M_i, W_p act on their named channels. The uncertainty-factor row/column orders are (z_S, z_G, w_S, w_G) for Θ_Δ and (w_S, w_G, z_S, z_G) for $\mathbf{D}(\Theta_\Delta)$. The performance-factor orders are (z_p, w_p) for Θ_p and (w_p, z_p) for $\mathbf{D}(\Theta_p)$, with the same orders for their corresponding filtered outputs.

Graph and signature blocks. In the primal graph, $(\mathcal{C}_z, \mathcal{D}_z)$ produce $\tilde{z}$ (5 rows), while $(\mathcal{C}_w, \mathcal{D}_w)$ produce $\tilde{w}$ (3 rows); both feedthrough blocks have input w (3 columns). In the dual graph, $(\underline{\mathcal{C}}_z, \underline{\mathcal{D}}_z)$ produce $\underline{\tilde{z}}$ (3 rows), while $(\underline{\mathcal{C}}_w, \underline{\mathcal{D}}_w)$ produce $\underline{\tilde{w}}$ (5 rows); both feedthrough blocks have input $\underline{w}$ (5 columns). The primal signature V uses the primal filtered order $(\tilde{z}_\Delta, \tilde{z}_p, \tilde{w}_\Delta, \tilde{w}_p)$. The dual signatures $\underline{V}$ use $(\underline{\tilde{z}}_\Delta, \underline{\tilde{z}}_p, \underline{\tilde{w}}_\Delta, \underline{\tilde{w}}_p)$.

Generalized plant operator tuple

$\mathcal{T}_P$

dimensions [2; 3] $\leftarrow$ [2; 3]

$$\mathcal{T}_P = \mathcal{P} \left[\begin{array}{c|c} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ \hline \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} & \left\{ \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \right\} \end{array} \right]$$

$\mathcal{A}_P$

dimensions [2; 3] $\leftarrow$ [2; 3]

$$\mathcal{A}_P = \mathcal{P} \left[\begin{array}{c|c} \begin{bmatrix} -166.67 & 0 \\ 0 & -71.43 \end{bmatrix} & \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ \hline \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} & \left\{ \begin{bmatrix} -166.67 & 0 & 0 \\ 0 & -166.67 & 0 \\ 0 & 0 & -250 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \right\} \end{array} \right]$$

$\mathcal{B}_P$

dimensions [2; 3] $\leftarrow$ [3; 0]

$$\mathcal{B}_P = \mathcal{P} \left[\begin{array}{c|c} \begin{bmatrix} 166.67 & 0 & 1 \\ 0 & 71.43 & 0 \end{bmatrix} & 0 \\ \hline \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} & \{0, 0, 0\} \end{array} \right]$$

$\mathcal{B}_u$

dimensions [2; 3] $\leftarrow$ [1; 0]

$$\mathcal{B}_u = \mathcal{P} \left[\begin{array}{c|c} \begin{bmatrix} 4600 \\ 0 \end{bmatrix} & 0 \\ \hline \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} & \{0, 0, 0\} \end{array} \right]$$

$\mathcal{C}_P$

dimensions [5; 0] $\leftarrow$ [2; 3]

$$\mathcal{C}_P = \mathcal{P} \left[\begin{array}{c|c} \begin{bmatrix} 0 & -10.7 \\ 20 & -12.3 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} & \begin{bmatrix} -10.7 & 0 & 0 \\ 0 & 20 & -12.3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ \hline 0 & \{0, 0, 0\} \end{array} \right]$$

$\mathcal{D}_P$

dimensions [5; 0] $\leftarrow$ [3; 0]

$$\mathcal{D}_P = \mathcal{P} \left[\begin{array}{c|c} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} & 0 \\ \hline 0 & \{0, 0, 0\} \end{array} \right]$$

$\mathcal{D}_{zu}$

dimensions [5; 0] $\leftarrow$ [1; 0]

$$\mathcal{D}_{zu} = \mathcal{P} \left[\begin{array}{c|c} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0.01 \end{bmatrix} & 0 \\ \hline 0 & \{0, 0, 0\} \end{array} \right]$$

Laplace-domain multiplier and performance filters

$$H_S(s)$$

$$H_S(s) = \frac{144.56 s^2 + 4.28 \times 10^4 s + 2.58 \times 10^6}{s^3 + 488.1 s^2 + 7.14 \times 10^4 s + 2.98 \times 10^6}$$

$$M_S(s)$$

$$M_S(s) = \frac{s^3 + 343.53 s^2 + 2.87 \times 10^4 s + 3.97 \times 10^5}{s^3 + 488.1 s^2 + 7.14 \times 10^4 s + 2.98 \times 10^6}$$

$$H_G(s)$$

$$H_G(s) = \frac{144.56 s^2 + 4.28 \times 10^4 s + 2.58 \times 10^6}{s^3 + 488.1 s^2 + 7.14 \times 10^4 s + 2.98 \times 10^6}$$

$$M_G(s)$$

$$M_G(s) = \frac{s^3 + 343.53 s^2 + 2.87 \times 10^4 s + 3.97 \times 10^5}{s^3 + 488.1 s^2 + 7.14 \times 10^4 s + 2.98 \times 10^6}$$

$$\Theta_\Delta(s)$$

$$\Theta_\Delta(s) = \begin{bmatrix} \frac{-0.27 s^3 - 93.63 s^2 - 7810.76 s - 1.08 \times 10^5}{s^3 + 488.1 s^2 + 7.14 \times 10^4 s + 2.98 \times 10^6} & 0 & \frac{-0.05 s^3 - 124.12 s^2 - 3.29 \times 10^4 s - 1.91 \times 10^6}{s^3 + 488.1 s^2 + 7.14 \times 10^4 s + 2.98 \times 10^6} & 0 \\ 0 & \frac{-0.27 s^3 - 93.63 s^2 - 7810.76 s - 1.08 \times 10^5}{s^3 + 488.1 s^2 + 7.14 \times 10^4 s + 2.98 \times 10^6} & 0 & \frac{-0.05 s^3 - 124.12 s^2 - 3.29 \times 10^4 s - 1.91 \times 10^6}{s^3 + 488.1 s^2 + 7.14 \times 10^4 s + 2.98 \times 10^6} \\ \frac{-0.27 s^3 - 93.63 s^2 - 7810.75 s - 1.08 \times 10^5}{s^3 + 488.1 s^2 + 7.14 \times 10^4 s + 2.98 \times 10^6} & 0 & \frac{1.42 s^3 + 592.25 s^2 + 7.19 \times 10^4 s + 2.45 \times 10^6}{s^3 + 488.1 s^2 + 7.14 \times 10^4 s + 2.98 \times 10^6} & 0 \\ 0 & \frac{-0.27 s^3 - 93.63 s^2 - 7810.75 s - 1.08 \times 10^5}{s^3 + 488.1 s^2 + 7.14 \times 10^4 s + 2.98 \times 10^6} & 0 & \frac{1.42 s^3 + 592.25 s^2 + 7.19 \times 10^4 s + 2.45 \times 10^6}{s^3 + 488.1 s^2 + 7.14 \times 10^4 s + 2.98 \times 10^6} \end{bmatrix}$$

$$\mathbf{D}(\Theta_\Delta)(s)$$

$$\mathbf{D}(\Theta_\Delta)(s) = \begin{bmatrix} \frac{0.68 s^3 + 234.06 s^2 + 1.95 \times 10^4 s + 2.70 \times 10^5}{s^3 + 343.53 s^2 + 2.87 \times 10^4 s + 3.97 \times 10^5} & 0 & \frac{0.13 s^3 + 310.29 s^2 + 8.22 \times 10^4 s + 4.78 \times 10^6}{s^3 + 343.53 s^2 + 2.87 \times 10^4 s + 3.97 \times 10^5} & 0 \\ 0 & \frac{0.68 s^3 + 234.06 s^2 + 1.95 \times 10^4 s + 2.70 \times 10^5}{s^3 + 343.53 s^2 + 2.87 \times 10^4 s + 3.97 \times 10^5} & 0 & \frac{0.13 s^3 + 310.29 s^2 + 8.22 \times 10^4 s + 4.78 \times 10^6}{s^3 + 343.53 s^2 + 2.87 \times 10^4 s + 3.97 \times 10^5} \\ \frac{0.68 s^3 + 234.06 s^2 + 1.95 \times 10^4 s + 2.70 \times 10^5}{s^3 + 343.53 s^2 + 2.87 \times 10^4 s + 3.97 \times 10^5} & 0 & \frac{-3.54 s^3 - 1480.62 s^2 - 1.80 \times 10^5 s - 6.14 \times 10^6}{s^3 + 343.53 s^2 + 2.87 \times 10^4 s + 3.97 \times 10^5} & 0 \\ 0 & \frac{0.68 s^3 + 234.06 s^2 + 1.95 \times 10^4 s + 2.70 \times 10^5}{s^3 + 343.53 s^2 + 2.87 \times 10^4 s + 3.97 \times 10^5} & 0 & \frac{-3.54 s^3 - 1480.62 s^2 - 1.80 \times 10^5 s - 6.14 \times 10^6}{s^3 + 343.53 s^2 + 2.87 \times 10^4 s + 3.97 \times 10^5} \end{bmatrix}$$

$$W_p(s)$$

$$W_p(s) = 1$$

$$W_p^{-1}(s)$$

$$W_p^{-1}(s) = 1$$

$$\Theta_p(s)$$

$$\Theta_p(s) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{D}(\Theta_p)(s)$$

$$\mathbf{D}(\Theta_p)(s) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\Theta(s)$$

$$\Theta(s) = \begin{bmatrix} \frac{-0.27 s^3 - 93.63 s^2 - 7810.76 s - 1.08 \times 10^5}{s^3 + 488.1 s^2 + 7.14 \times 10^4 s + 2.98 \times 10^6} & 0 & 0 & 0 & 0 & \frac{-0.05 s^3 - 124.12 s^2 - 3.29 \times 10^4 s - 1.91 \times 10^6}{s^3 + 488.1 s^2 + 7.14 \times 10^4 s + 2.98 \times 10^6} & 0 & 0 \\ 0 & \frac{-0.27 s^3 - 93.63 s^2 - 7810.76 s - 1.08 \times 10^5}{s^3 + 488.1 s^2 + 7.14 \times 10^4 s + 2.98 \times 10^6} & 0 & 0 & 0 & 0 & \frac{-0.05 s^3 - 124.12 s^2 - 3.29 \times 10^4 s - 1.91 \times 10^6}{s^3 + 488.1 s^2 + 7.14 \times 10^4 s + 2.98 \times 10^6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ \frac{-0.27 s^3 - 93.63 s^2 - 7810.75 s - 1.08 \times 10^5}{s^3 + 488.1 s^2 + 7.14 \times 10^4 s + 2.98 \times 10^6} & 0 & 0 & 0 & 0 & \frac{1.42 s^3 + 592.25 s^2 + 7.19 \times 10^4 s + 2.45 \times 10^6}{s^3 + 488.1 s^2 + 7.14 \times 10^4 s + 2.98 \times 10^6} & 0 & 0 \\ 0 & \frac{-0.27 s^3 - 93.63 s^2 - 7810.75 s - 1.08 \times 10^5}{s^3 + 488.1 s^2 + 7.14 \times 10^4 s + 2.98 \times 10^6} & 0 & 0 & 0 & 0 & \frac{1.42 s^3 + 592.25 s^2 + 7.19 \times 10^4 s + 2.45 \times 10^6}{s^3 + 488.1 s^2 + 7.14 \times 10^4 s + 2.98 \times 10^6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{D}(\Theta)(s)$$

$$\mathbf{D}(\Theta)(s) = \begin{bmatrix} \frac{0.68 s^3 + 234.06 s^2 + 1.95 \times 10^4 s + 2.70 \times 10^5}{s^3 + 343.53 s^2 + 2.87 \times 10^4 s + 3.97 \times 10^5} & 0 & 0 & \frac{0.13 s^3 + 310.29 s^2 + 8.22 \times 10^4 s + 4.78 \times 10^6}{s^3 + 343.53 s^2 + 2.87 \times 10^4 s + 3.97 \times 10^5} & 0 & 0 & 0 & 0 \\ 0 & \frac{0.68 s^3 + 234.06 s^2 + 1.95 \times 10^4 s + 2.70 \times 10^5}{s^3 + 343.53 s^2 + 2.87 \times 10^4 s + 3.97 \times 10^5} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ \frac{0.68 s^3 + 234.06 s^2 + 1.95 \times 10^4 s + 2.70 \times 10^5}{s^3 + 343.53 s^2 + 2.87 \times 10^4 s + 3.97 \times 10^5} & 0 & 0 & \frac{-3.54 s^3 - 1480.62 s^2 - 1.80 \times 10^5 s - 6.14 \times 10^6}{s^3 + 343.53 s^2 + 2.87 \times 10^4 s + 3.97 \times 10^5} & 0 & 0 & 0 & 0 \\ 0 & \frac{0.68 s^3 + 234.06 s^2 + 1.95 \times 10^4 s + 2.70 \times 10^5}{s^3 + 343.53 s^2 + 2.87 \times 10^4 s + 3.97 \times 10^5} & 0 & 0 & \frac{-3.54 s^3 - 1480.62 s^2 - 1.80 \times 10^5 s - 6.14 \times 10^6}{s^3 + 343.53 s^2 + 2.87 \times 10^4 s + 3.97 \times 10^5} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Recovered controller

$$\mathcal{K} = [\mathcal{K}_P \ \mathcal{K}_\Theta]$$

dimensions [1; 0] $\leftarrow$ [8; 3]

$$\mathcal{K} = [\mathcal{K}_P \ \mathcal{K}_\Theta] = \mathcal{P} \left[\begin{array}{c|c} \begin{bmatrix} -0.18 & 0.13 & 0.06 & -0.38 & -0.57 & 5.22 \times 10^{-3} & -0.12 & -0.26 \end{bmatrix} & \begin{bmatrix} 5.30 \times 10^{-6} s^{12} - 0.01 s^{11} + 0.05 s^{10} - 0.04 s^9 & -0.03 s^{12} + 0.48 s^{11} - 2.31 s^{10} + 5.44 s^9 & 0.36 s^{12} - 2.62 s^{11} + 8.5 s^{10} - 15 s^9 \\ -0.13 s^8 + 0.36 s^7 - 0.31 s^6 - 0.01 s^5 & -6.72 s^8 + 3.49 s^7 + 1.31 s^6 - 3.39 s^5 & +14.27 s^8 - 5.08 s^7 - 2.79 s^6 + 3.84 s^5 \\ +0.2 s^4 - 0.13 s^3 + 0.04 s^2 - 0.06 s & +2.99 s^4 - 1.42 s^3 - 0.92 s^2 + 0.92 s & -1.94 s^4 - 0.2 s^3 + 0.75 s^2 - 7.14 \times 10^{-3} s \\ +0.12 & +0.12 & -0.17 \end{bmatrix} \\ \hline 0 & \{0, 0, 0\} \end{array} \right]$$

Primal augmented graph $\Theta[(K \star P); I]$

$\mathcal{T}$

dimensions [8; 3] $\leftarrow$ [8; 3]

$$\mathcal{T} = \mathcal{P} \left[\left[\begin{array}{cccccccc|cccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{array} \right] \middle| \begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \\ \left[\begin{array}{cccccccc|cccc} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \middle| \left\{ \begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right\}, \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right], \left[\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right] \right\} \right]$$

$\mathcal{A}$

dimensions [8; 3] $\leftarrow$ [8; 3]

$$\mathcal{A} = \mathcal{P} \left[\left[\begin{array}{cccccccc} -993.07 & 588.07 & 283.75 & -1734.27 & -2616.65 & 24.02 & -539.31 & -1191.29 \\ 0 & -71.43 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -14.21 & -250 & 268.72 & 184.38 & 0 & 0 & 0 \\ 0 & -0.57 & 0 & -166.67 & -147.56 & 0 & 0 & 0 \\ 0 & 24.57 & 0 & 0 & -71.43 & 0 & 0 & 0 \\ 26.56 & -16.34 & 0 & 0 & 0 & -250 & 268.72 & 184.38 \\ 1.06 & -0.65 & 0 & 0 & 0 & 0 & -166.67 & -147.56 \\ -45.92 & 28.24 & 0 & 0 & 0 & 0 & 0 & -71.43 \end{array} \right] \middle| \left[\begin{array}{cccc} 0.02 s^{12} - 56.91 s^{11} + 208.15 s^{10} - 175.02 s^9 & -147.22 s^{12} + 2226.74 s^{11} - 1.06 \times 10^4 s^{10} + 2.50 \times 10^4 s^9 & 1660.07 s^{12} - 1.21 \times 10^4 s^{11} + 3.91 \times 10^4 s^{10} - 6.90 \times 10^4 s^9 & 0 \\ -594.96 s^8 + 1647.33 s^7 - 1411.34 s^6 - 57.41 s^5 & -3.09 \times 10^4 s^8 + 1.61 \times 10^4 s^7 + 6017.06 s^6 - 1.56 \times 10^4 s^5 & +6.57 \times 10^4 s^8 - 2.34 \times 10^4 s^7 - 1.28 \times 10^4 s^6 + 1.77 \times 10^4 s^5 & 0 \\ +908.93 s^4 - 577.84 s^3 + 187.81 s^2 - 273.97 s & +1.38 \times 10^4 s^4 - 6550.99 s^3 - 4223.47 s^2 + 4246.7 s & -8910.33 s^4 - 934.22 s^3 + 3461.58 s^2 - 32.83 s & 0 \\ +546.44 & +549.01 & -789.1 & 0 \\ 0 & 0 & 0 & 0 \\ -14.21 & 0 & 0 & 0 \\ -0.57 & 0 & 0 & 0 \\ 24.57 & 0 & 0 & 0 \\ 0 & 26.56 & 0 & -16.34 \\ 0 & 1.06 & 0 & -0.65 \\ 0 & -45.92 & 0 & 28.24 \end{array} \right] \\ \left[\begin{array}{cccccccc} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \middle| \left\{ \begin{array}{ccc} -166.67 & 0 & 0 \\ 0 & -166.67 & 0 \\ 0 & 0 & -250 \end{array} \right\}, \left[\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right], \left[\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right] \right\} \right]$$

$\mathcal{B}$

dimensions [8; 3] $\leftarrow$ [3; 0]

$$\mathcal{B} = \mathcal{P} \left[\left[\begin{array}{ccc} 166.67 & 0 & 1 \\ 0 & 71.43 & 0 \\ -3.32 & 0 & 0 \\ -0.13 & 0 & 0 \\ 5.74 & 0 & 0 \\ 0 & -3.32 & 0 \\ 0 & -0.13 & 0 \\ 0 & 5.74 & 0 \end{array} \right] \middle| 0 \right] \\ \left[\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right] \middle| \{0, 0, 0\} \right]$$

$\mathcal{C}_z$

dimensions [5; 0] $\leftarrow$ [8; 3]

$$\mathcal{C}_z = \mathcal{P} \left[\left[\begin{array}{cccccccc} 0 & 2.92 & 1.75 & -10.87 & -16.4 & 0 & 0 & 0 \\ -5.45 & 3.35 & 0 & 0 & 0 & 1.75 & -10.87 & -16.4 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1.80 \times 10^{-3} & 1.28 \times 10^{-3} & 6.17 \times 10^{-4} & -3.77 \times 10^{-3} & -5.69 \times 10^{-3} & 5.22 \times 10^{-5} & -1.17 \times 10^{-3} & -2.59 \times 10^{-3} \end{array} \right] \middle| \left[\begin{array}{cccc} 2.92 & 0 & 0 & 0 \\ 0 & -5.45 & 3.35 & 0 \\ 0 & 0 & 0 & 0 \\ 5.30 \times 10^{-8} s^{12} - 1.24 \times 10^{-4} s^{11} + 4.53 \times 10^{-4} s^{10} - 3.80 \times 10^{-4} s^9 & -3.20 \times 10^{-4} s^{12} + 4.84 \times 10^{-3} s^{11} - 0.02 s^{10} + 0.05 s^9 & 3.61 \times 10^{-3} s^{12} - 0.03 s^{11} + 0.08 s^{10} - 0.15 s^9 & 0 \\ -1.29 \times 10^{-3} s^8 + 3.58 \times 10^{-3} s^7 - 3.07 \times 10^{-3} s^6 - 1.25 \times 10^{-4} s^5 & -0.07 s^8 + 0.03 s^7 + 0.01 s^6 - 0.03 s^5 & +0.14 s^8 - 0.05 s^7 - 0.03 s^6 + 0.04 s^5 & 0 \\ +1.98 \times 10^{-3} s^4 - 1.26 \times 10^{-3} s^3 + 4.08 \times 10^{-4} s^2 - 5.96 \times 10^{-4} s & +0.03 s^4 - 0.01 s^3 - 9.18 \times 10^{-3} s^2 + 9.23 \times 10^{-3} s & -0.02 s^4 - 2.03 \times 10^{-3} s^3 + 7.53 \times 10^{-3} s^2 - 7.14 \times 10^{-5} s & 0 \\ +1.19 \times 10^{-3} & +1.19 \times 10^{-3} & -1.72 \times 10^{-3} & 0 \end{array} \right] \\ 0 \middle| \{0, 0, 0\} \right]$$

$\mathcal{C}_w$

dimensions [3; 0] $\leftarrow$ [8; 3]

$$\mathcal{C}_w = \mathcal{P} \left[\left[\begin{array}{cccccc} 0 & 2.92 & 1.75 & -10.87 & -16.4 & 0 \\ -5.45 & 3.35 & 0 & 0 & 0 & 1.75 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \middle| 0 \right] \left[\begin{array}{ccc} 2.92 & 0 & 0 \\ 0 & -5.45 & 3.35 \\ 0 & 0 & 0 \end{array} \right] \middle| \{0, 0, 0\} \right]$$

$\mathcal{D}_z$

dimensions [5; 0] $\leftarrow$ [3; 0]

$$\mathcal{D}_z = \mathcal{P} \left[\left[\begin{array}{ccc} -0.05 & 0 & 0 \\ 0 & -0.05 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right] \middle| 0 \right] \left[\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right] \middle| \{0, 0, 0\} \right]$$

$\mathcal{D}_w$

dimensions [3; 0] $\leftarrow$ [3; 0]

$$\mathcal{D}_w = \mathcal{P} \left[\left[\begin{array}{ccc} 1.42 & 0 & 0 \\ 0 & 1.42 & 0 \\ 0 & 0 & 1 \end{array} \right] \middle| 0 \right] \left[\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right] \middle| \{0, 0, 0\} \right]$$

Dual augmented graph $D(\Theta)[(K \star P)^*; I]$

$\mathcal{I}_{\text{cl}}$

dimensions [8; 3] $\leftarrow$ [8; 3]

$$\mathcal{I}_{\text{cl}} = \mathcal{P} \left[\begin{array}{c|c} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} & \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ \hline \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} & \left\{ \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\} \end{array} \right]$$

$\mathcal{A}_{\text{cl}}$

dimensions [8; 3] $\leftarrow$ [8; 3]

$$\mathcal{A}_{\text{cl}} = \mathcal{P} \left[\begin{array}{c|c} \begin{bmatrix} -993.07 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 588.07 & -71.43 & 0 & 0 & 0 & 0 & 0 & 0 \\ 283.75 & 0 & -241.47 & 0.34 & -14.75 & 0 & 0 & 0 \\ -1734.27 & 0 & 215.74 & -168.78 & 91.59 & 0 & 0 & 0 \\ -2616.65 & 0 & 104.47 & -150.75 & 66.72 & 0 & 0 & 0 \\ 24.02 & -3.37 \times 10^{-5} & 0 & 0 & 0 & -241.47 & 0.34 & -14.75 \\ -539.31 & 2.38 \times 10^{-4} & 0 & 0 & 0 & 215.74 & -168.78 & 91.59 \\ -1191.29 & 4.05 \times 10^{-4} & 0 & 0 & 0 & 104.47 & -150.75 & 66.72 \end{bmatrix} & \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ \hline \begin{bmatrix} 0.02 s^{12} - 56.91 s^{11} + 208.15 s^{10} - 175.02 s^9 \\ - 594.96 s^8 + 1647.33 s^7 - 1411.34 s^6 - 57.41 s^5 \\ + 908.93 s^4 - 577.84 s^3 + 187.81 s^2 - 273.97 s \\ + 546.44 \\ - 147.22 s^{12} + 2226.74 s^{11} - 1.06 \times 10^4 s^{10} + 2.50 \times 10^4 s^9 \\ - 3.09 \times 10^4 s^8 + 1.61 \times 10^4 s^7 + 6017.06 s^6 - 1.56 \times 10^4 s^5 \\ + 1.38 \times 10^4 s^4 - 6550.99 s^3 - 4223.47 s^2 + 4246.7 s \\ + 549.01 \\ 1660.07 s^{12} - 1.21 \times 10^4 s^{11} + 3.91 \times 10^4 s^{10} - 6.90 \times 10^4 s^9 \\ + 6.57 \times 10^4 s^8 - 2.34 \times 10^4 s^7 - 1.28 \times 10^4 s^6 + 1.77 \times 10^4 s^5 \\ - 8910.33 s^4 - 934.22 s^3 + 3461.58 s^2 - 32.83 s \\ - 789.1 \end{bmatrix} & \left\{ \begin{bmatrix} -166.67 & 0 & 0 \\ 0 & -166.67 & 0 \\ 0 & 0 & -250 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \right\} \end{array} \right]$$

$\mathcal{B}_{\text{cl}}$

dimensions [8; 3] $\leftarrow$ [5; 0]

$$\mathcal{B}_{\text{cl}} = \mathcal{P} \left[\begin{array}{c|c} \begin{bmatrix} 0 & 20 & 1 & 0 & -1.80 \times 10^{-3} \\ -10.7 & -12.3 & 0 & 1 & 1.28 \times 10^{-3} \\ -6.42 & 0 & 0 & 0 & 6.17 \times 10^{-4} \\ 39.89 & 0 & 0 & 0 & -3.77 \times 10^{-3} \\ 60.17 & 0 & 0 & 0 & -5.69 \times 10^{-3} \\ 0 & -6.42 & 0 & 0 & 5.22 \times 10^{-5} \\ 0 & 39.89 & 0 & 0 & -1.17 \times 10^{-3} \\ 0 & 60.17 & 0 & 0 & -2.59 \times 10^{-3} \end{bmatrix} & 0 \\ \hline \begin{bmatrix} 5.30 \times 10^{-8} s^{12} - 1.24 \times 10^{-4} s^{11} + 4.53 \times 10^{-4} s^{10} - 3.80 \times 10^{-4} s^9 \\ - 1.29 \times 10^{-3} s^8 + 3.58 \times 10^{-3} s^7 - 3.07 \times 10^{-3} s^6 - 1.25 \times 10^{-4} s^5 \\ + 1.98 \times 10^{-3} s^4 - 1.26 \times 10^{-3} s^3 + 4.08 \times 10^{-4} s^2 - 5.96 \times 10^{-4} s \\ + 1.19 \times 10^{-3} \\ - 3.20 \times 10^{-4} s^{12} + 4.84 \times 10^{-3} s^{11} - 0.02 s^{10} + 0.05 s^9 \\ - 0.07 s^8 + 0.03 s^7 + 0.01 s^6 - 0.03 s^5 \\ + 0.03 s^4 - 0.01 s^3 - 9.18 \times 10^{-3} s^2 + 9.23 \times 10^{-3} s \\ + 1.19 \times 10^{-3} \\ 3.61 \times 10^{-3} s^{12} - 0.03 s^{11} + 0.08 s^{10} - 0.15 s^9 \\ + 0.14 s^8 - 0.05 s^7 - 0.03 s^6 + 0.04 s^5 \\ - 0.02 s^4 - 2.03 \times 10^{-3} s^3 + 7.53 \times 10^{-3} s^2 - 7.14 \times 10^{-5} s \\ - 1.72 \times 10^{-3} \end{bmatrix} & \{0, 0, 0\} \end{array} \right]$$

$\mathcal{C}_z$

dimensions [3; 0] $\leftarrow$ [8; 3]

$$\mathcal{C}_z = \mathcal{P} \left[\begin{array}{c|c} \begin{bmatrix} 113.56 & 0 & -2.44 & -0.1 & 4.21 & 0 & 0 & 0 & 0 \\ 0 & 48.67 & 0 & 0 & 0 & -2.44 & -0.1 & 4.21 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} & \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ \hline 0 & \{0, 0, 0\} \end{array} \right]$$

$\mathcal{C}_w$

dimensions [5; 0] $\leftarrow$ [8; 3]

$$\mathcal{C}_w = \mathcal{P} \left[\begin{array}{c|c} \begin{bmatrix} 113.56 & 0 & 2.44 & 0.1 & -4.21 & 0 & 0 & 0 & 0 \\ 0 & 48.67 & 0 & 0 & 0 & 2.44 & 0.1 & -4.21 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} & \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ \hline 0 & \{0, 0, 0\} \end{array} \right]$$

$\mathcal{D}_z$

dimensions [3; 0] $\leftarrow$ [5; 0]

$$\mathcal{D}_z = \mathcal{P} \left[\begin{array}{c|c} \begin{bmatrix} 0.13 & 0 & 0 & 0 & 0 \\ 0 & 0.13 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} & 0 \\ \hline 0 & \{0, 0, 0\} \end{array} \right]$$

$$\underline{\mathcal{D}}_w = \mathcal{P} \left[\left[\begin{array}{ccccc|c} -3.54 & 0 & 0 & 0 & 0 & \\ 0 & -3.54 & 0 & 0 & 0 & \\ 0 & 0 & 1 & 0 & 0 & \\ 0 & 0 & 0 & 1 & 0 & \\ 0 & 0 & 0 & 0 & 1 & \end{array} \right] \begin{array}{c} 0 \\ \\ \\ \\ \end{array} \right]$$

0

$\{0, 0, 0\}$

Solved performance signatures

$\underline{V}_{\text{syn}} = \begin{bmatrix} 9.49 \times 10^{-4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3.14 \times 10^{-4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -9.49 \times 10^{-4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -3.14 \times 10^{-4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1.82 \times 10^{-4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1.82 \times 10^{-4} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1.82 \times 10^{-4} \end{bmatrix}$

$V_{\text{ana}} = \begin{bmatrix} 1.23 \times 10^7 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2.45 \times 10^7 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1.23 \times 10^7 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -2.45 \times 10^7 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2.65 \times 10^8 \end{bmatrix}$

$\underline{V}_{\text{ana}} = \begin{bmatrix} 1.28 \times 10^{-3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2.55 \times 10^{-4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1.28 \times 10^{-3} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2.55 \times 10^{-4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1.37 \times 10^{-4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1.37 \times 10^{-4} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1.37 \times 10^{-4} \end{bmatrix}$

Controller location. $\mathbf{x}_S$.